

PSTAT 126: REGRESSION ANALYSIS

Final Project

Part One

1. Selecting Random Sample

```
library(tidyverse)
library(patchwork)
```

```
set.seed(124)
diamonds <- read.csv("Diamonds Prices2022.csv")
sample_diamonds <- diamonds |> sample_n(1000)
summary(sample_diamonds)
```

```
##           X           carat           cut           color
## Min.      : 17   Min.      :0.2300   Length:1000   Length:1000
## 1st Qu.:13238   1st Qu.:0.4000   Class :character   Class :character
## Median :26364   Median :0.7100   Mode  :character   Mode  :character
## Mean    :26866   Mean    :0.8018
## 3rd Qu.:40670   3rd Qu.:1.0400
## Max.    :53935   Max.    :3.0000
##   clarity      depth      table      price
## Length:1000   Min.      :53.10   Min.      :53.00   Min.      : 348
## Class :character 1st Qu.:61.10   1st Qu.:56.00   1st Qu.: 958
## Mode  :character Median :61.80   Median :57.00   Median : 2441
##                Mean  :61.75   Mean  :57.53   Mean  : 3898
##                3rd Qu.:62.50   3rd Qu.:59.00   3rd Qu.: 5373
##                Max.   :68.30   Max.   :68.00   Max.   :18524
##           x           y           z
## Min.      :3.920   Min.      :3.950   Min.      :2.410
## 1st Qu.:4.740   1st Qu.:4.750   1st Qu.:2.947
## Median :5.700   Median :5.720   Median :3.540
## Mean    :5.747   Mean    :5.747   Mean    :3.548
## 3rd Qu.:6.540   3rd Qu.:6.540   3rd Qu.:4.030
## Max.    :9.040   Max.    :8.980   Max.    :5.970
```

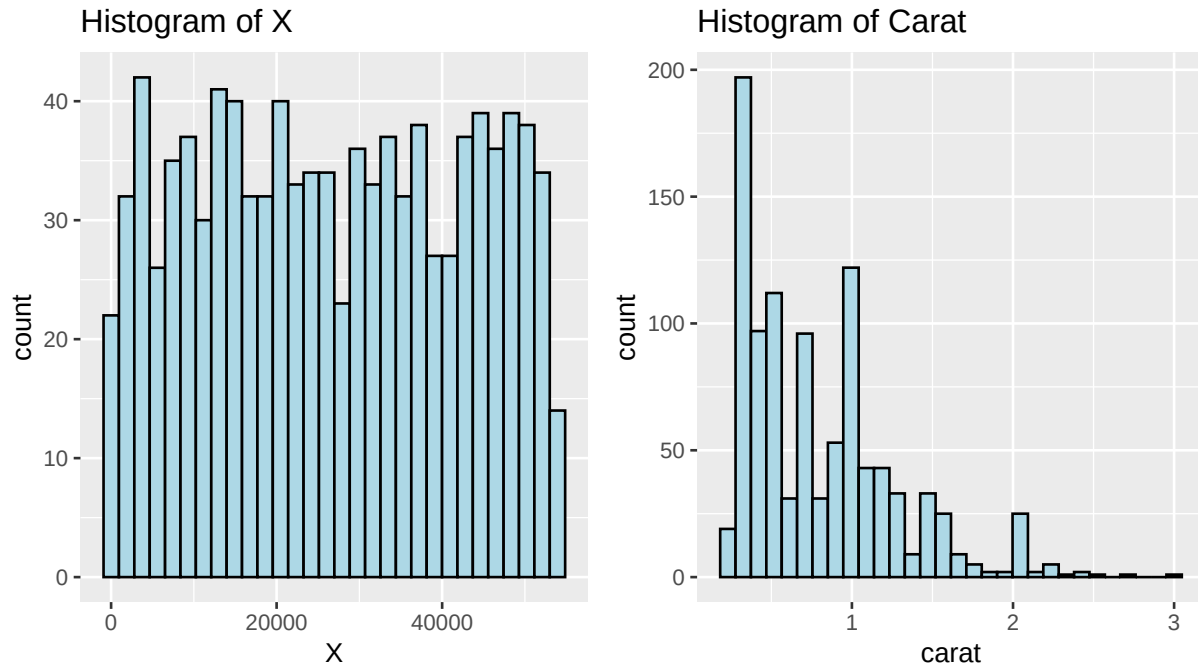
2) Description of Variables

```
p1 <- ggplot(sample_diamonds, aes(x = X)) +
  geom_histogram(bins = 30, fill = "lightblue", color = "black") +
  labs(title = "Histogram of X")

p2 <- ggplot(sample_diamonds, aes(x = carat)) +
  geom_histogram(bins = 30, fill = "lightblue", color = "black") +
```

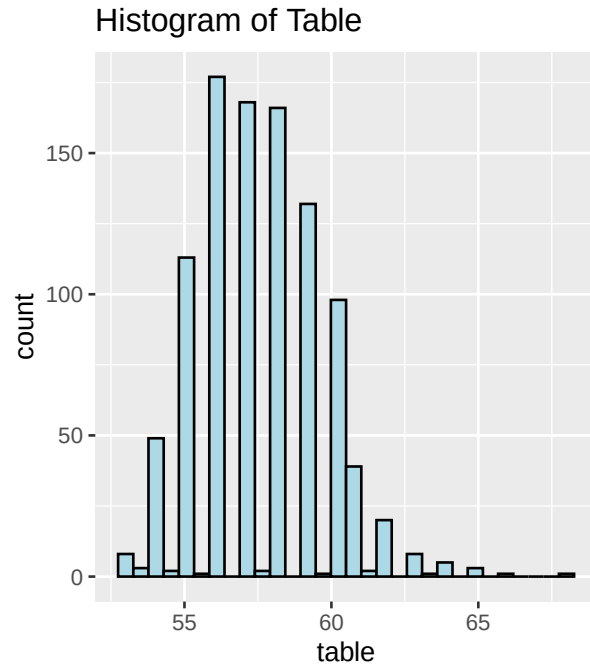
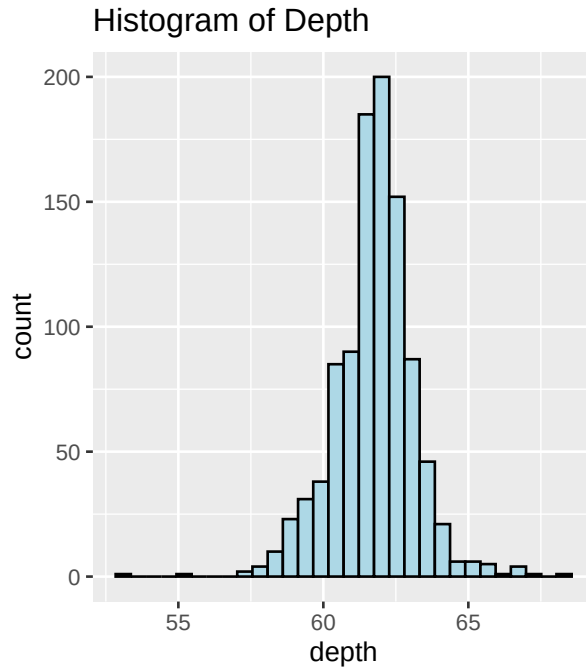
```
labs(title = "Histogram of Carat")
```

p1 | p2



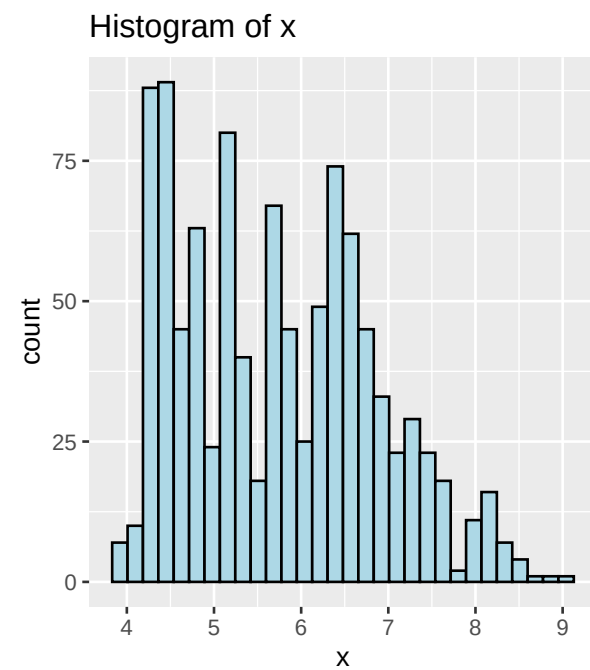
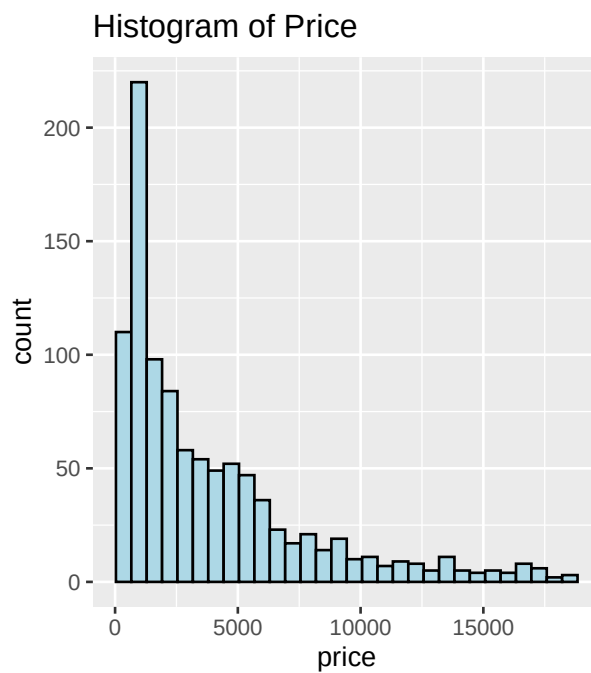
XVariable: The histogram for x appears approximately uniform, as evidenced by its relatively rectangular shape and the absence of pronounced peaks or clustering.

Carat: The distribution of carat is left-skewed with a long, gradually decreasing tail as carat values increase. This pattern suggests that a transformation (e.g., log or square root) may be appropriate if normalization is desired.



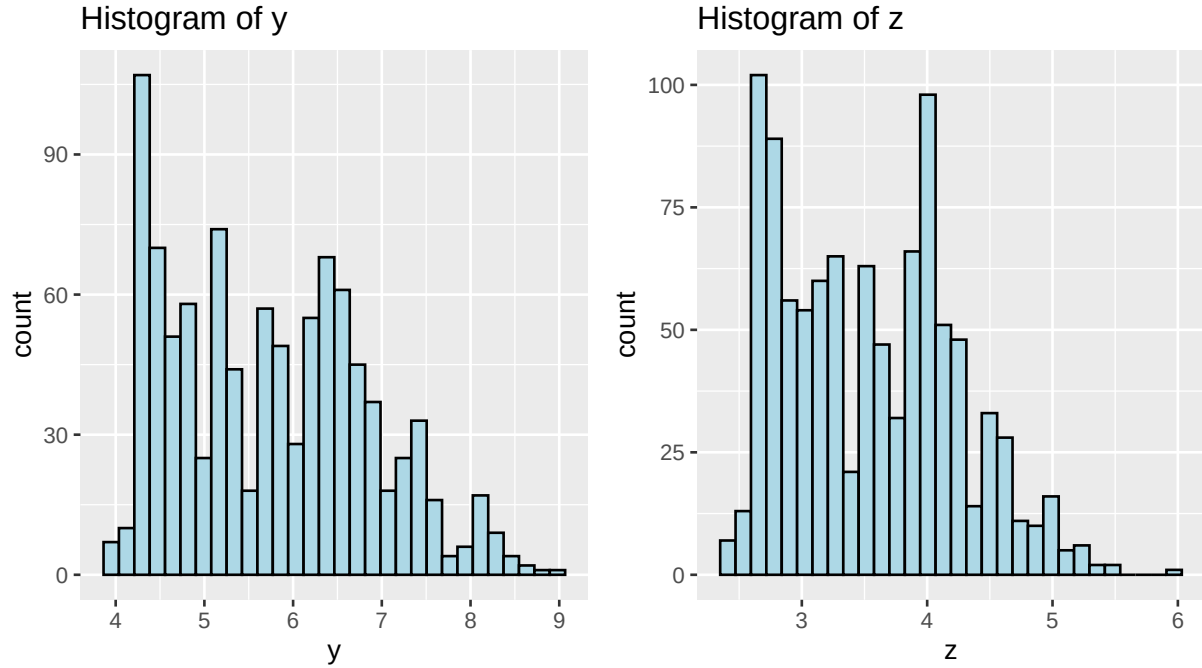
Depth: In the random sample, depth displays a shape that is close to normal, characterized by a narrow central peak. This suggests that the variable may reasonably follow a normal distribution.

Table Variable: The histogram for table resembles a near-normal distribution; however, noticeable left skewness and substantial gaps between some bins indicate deviations from perfect normality.



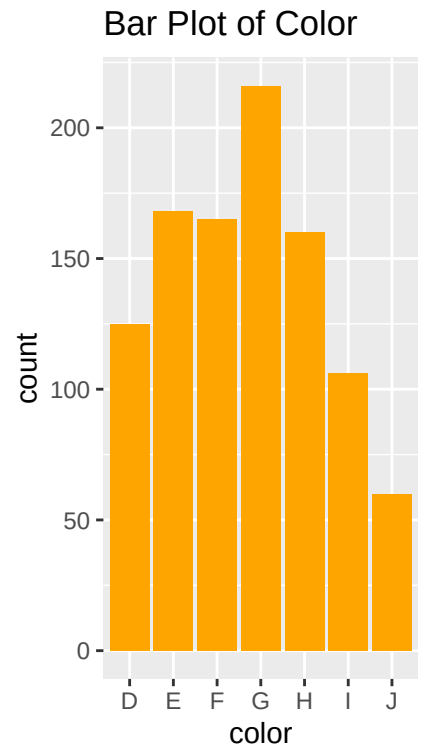
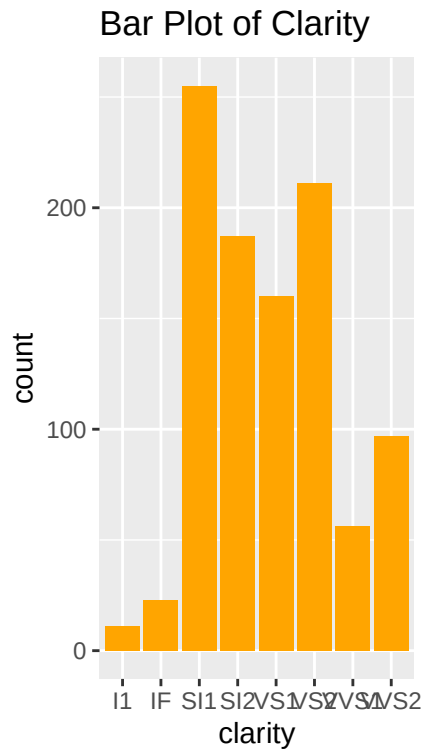
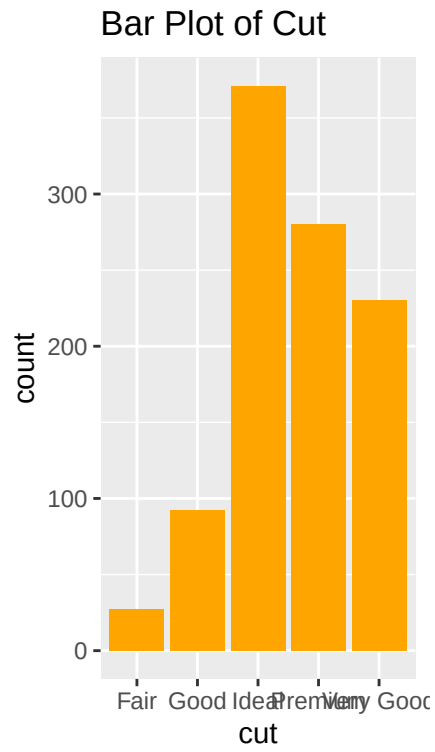
Price: The price distribution is strongly left-skewed, showing a steep decline as values increase. Its shape closely parallels the distribution of carat, again suggesting a potential need for transformation depending on modeling goals.

x (Length): The histogram for x (length) does not exhibit features clearly aligned with common theoretical distributions. Mild left skewness is present, but no dominant pattern emerges.



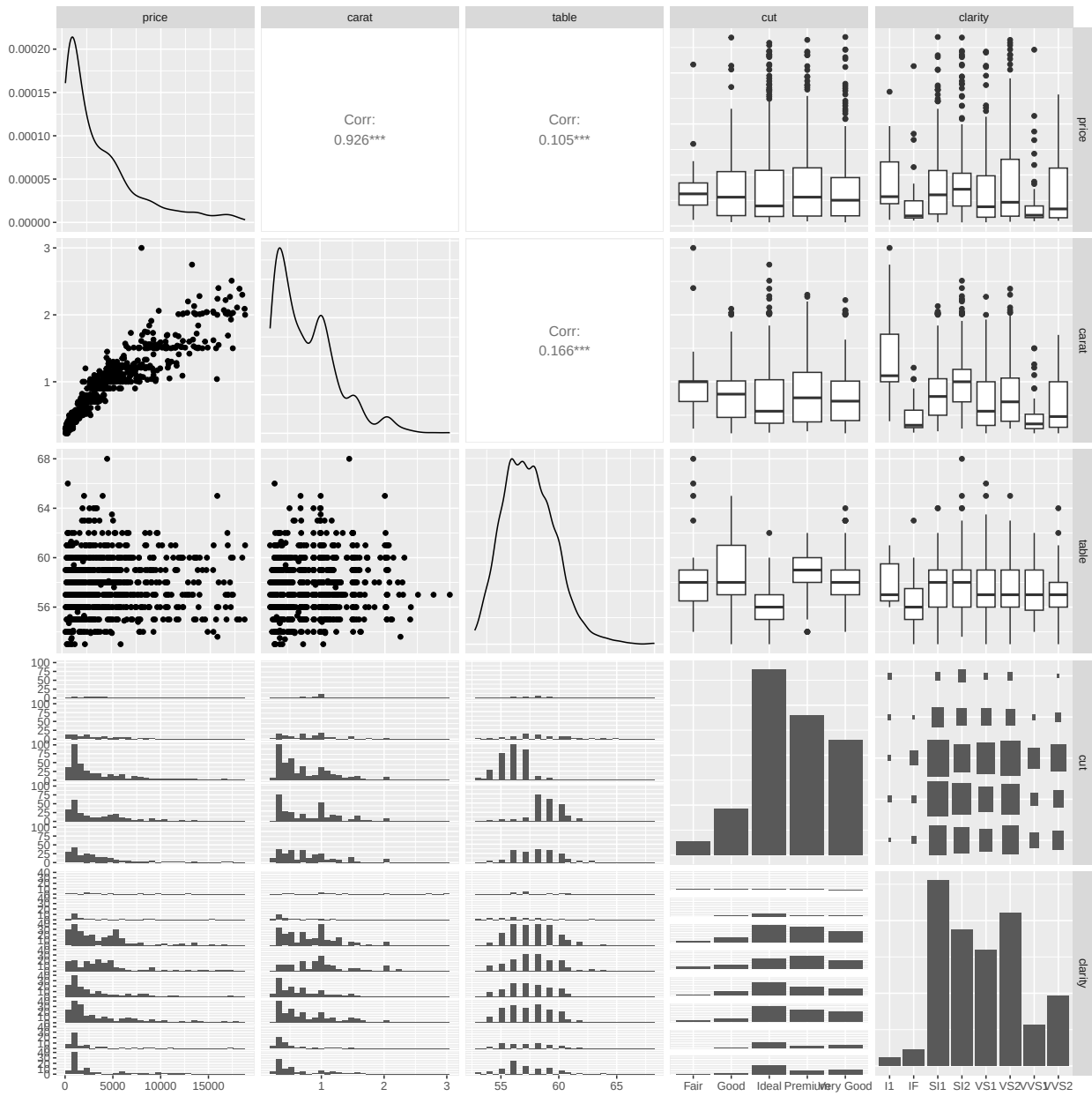
Y (Width): The y (width) variable also lacks resemblance to standard distributions and displays an uneven central region, indicating imbalance in the frequency of mid-range values.

Z (Depth): The distribution of z mirrors that of y, showing no clear alignment with known distributional shapes and exhibiting similar irregularities.



3) Selection of Variables and Correlation (3 quantitative 2 categorical)

PRICE, CARAT, DEPTH, TABLE, CUT, CLARITY



4) Running the LRM with Summary Statistics

```
diamond_model <- lm(price ~ ., data = diamond_subset)
summary(diamond_model)

##
## Call:
## lm(formula = price ~ ., data = diamond_subset)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -8901.9  -584.4  -107.4   455.6  8572.2
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -5883.52    1356.47  -4.337 1.59e-05 ***
## carat         8252.67      89.11   92.614 < 2e-16 ***
## table        -33.83      22.35   -1.514 0.130406
## cutGood       1023.67     272.53    3.756 0.000183 ***
## cutIdeal      1319.25     256.84    5.136 3.38e-07 ***
## cutPremium    1131.81     251.85    4.494 7.82e-06 ***
## cutVery Good  1063.16     255.15    4.167 3.36e-05 ***
## clarityIF      5162.42     464.10   11.124 < 2e-16 ***
## claritySI1     3646.17     387.72    9.404 < 2e-16 ***
## claritySI2     2987.51     387.72    7.705 3.17e-14 ***
## clarityVS1     4448.59     394.05   11.289 < 2e-16 ***
## clarityVS2     4338.21     389.75   11.131 < 2e-16 ***
## clarityVVS1     4814.08     420.48   11.449 < 2e-16 ***
## clarityVVS2     4769.16     403.60   11.817 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1232 on 986 degrees of freedom
## Multiple R-squared:  0.9009, Adjusted R-squared:  0.8996
## F-statistic: 689.4 on 13 and 986 DF,  p-value: < 2.2e-16
```

5) Comments on the Summary statistics

The regression results align with expectations in showing that a one-carat increase corresponds to a substantial rise in price—approximately \$8,254—which reflects the strong economic value associated with larger diamonds. However, the magnitude of the effects for several appearance-related characteristics was more surprising. In particular, clarity exhibited a notably large influence on price, indicating that visual quality plays a critical role in market valuation. The significance of cut was also unexpected, suggesting that consumers and evaluators are highly sensitive to multiple characteristics of a diamond's appearance. Collectively, these findings imply that the diamond market is both strict and highly discerning. Additionally, the model's adjusted R^2 of 0.89 (approximately similar by .01 decimal point of R^2 of .9009) indicates that it explains a remarkably high proportion of the variability in price, demonstrating strong overall model fit.

Part 2

1. Choosing the Simple Model

```
simple_model1 <- lm(carat~clarity, data= diamond_subset)
simple_model1

##
## Call:
## lm(formula = carat ~ clarity, data = diamond_subset)
##
## Coefficients:
## (Intercept)    clarityIF    claritySI1    claritySI2    clarityVS1    clarityVS2
##      1.4009      -0.9022      -0.5531      -0.3872      -0.7007      -0.5989
## clarityVVS1  clarityVVS2
##      -0.9138      -0.7763
```

2. Running the Model and Summary Statistics

```
summary(simple_model1)

##
## Call:
## lm(formula = carat ~ clarity, data = diamond_subset)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.9909 -0.3184 -0.1020  0.2080  1.5991
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.4009     0.1337  10.476 < 2e-16 ***
## clarityIF     -0.9022     0.1626  -5.549 3.69e-08 ***
## claritySI1    -0.5531     0.1366  -4.049 5.54e-05 ***
## claritySI2    -0.3872     0.1376  -2.814 0.00499 **
## clarityVS1    -0.7007     0.1383  -5.068 4.78e-07 ***
## clarityVS2    -0.5989     0.1372  -4.366 1.40e-05 ***
## clarityVVS1   -0.9138     0.1463  -6.247 6.20e-10 ***
## clarityVVS2   -0.7763     0.1411  -5.501 4.80e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4435 on 992 degrees of freedom
## Multiple R-squared:  0.1145, Adjusted R-squared:  0.1083
## F-statistic: 18.33 on 7 and 992 DF,  p-value: < 2.2e-16
```

```
library(ggplot2)
library(dplyr)

clarity_levels <- data.frame(clarity = unique(diamond_subset$clarity))

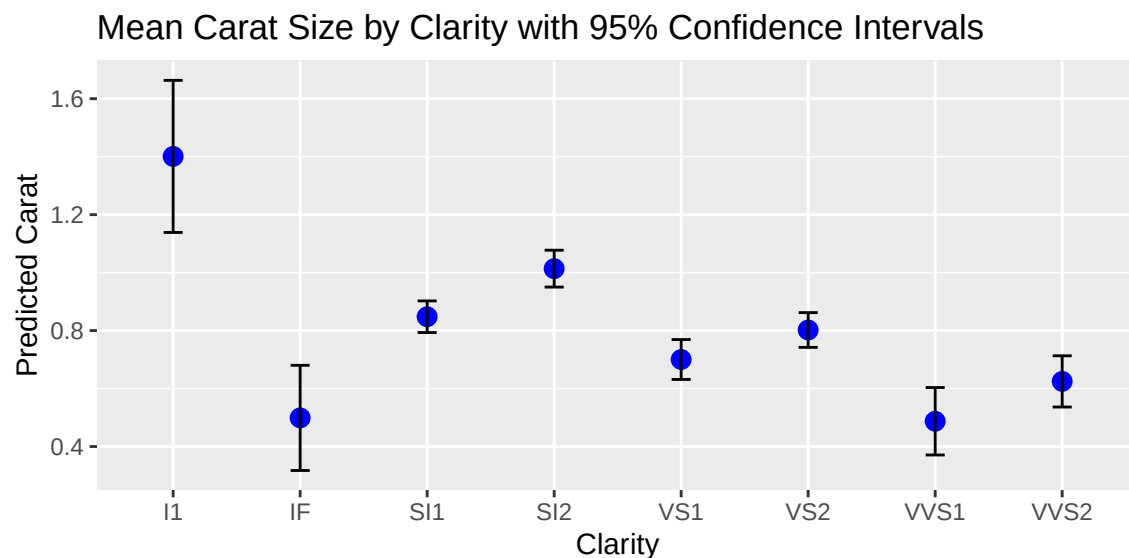
pred_df <- cbind(
```

```

    clarity_levels,
    predict(simple_model1, newdata = clarity_levels,
            interval = "confidence")
)

ggplot(pred_df, aes(x = clarity, y = fit)) +
  geom_point(size = 3, color = "blue") +
  geom_errorbar(aes(ymin = lwr, ymax = upr), width = 0.15) +
  labs(
    title = "Mean Carat Size by Clarity with 95% Confidence Intervals",
    x = "Clarity",
    y = "Predicted Carat"
  )

```

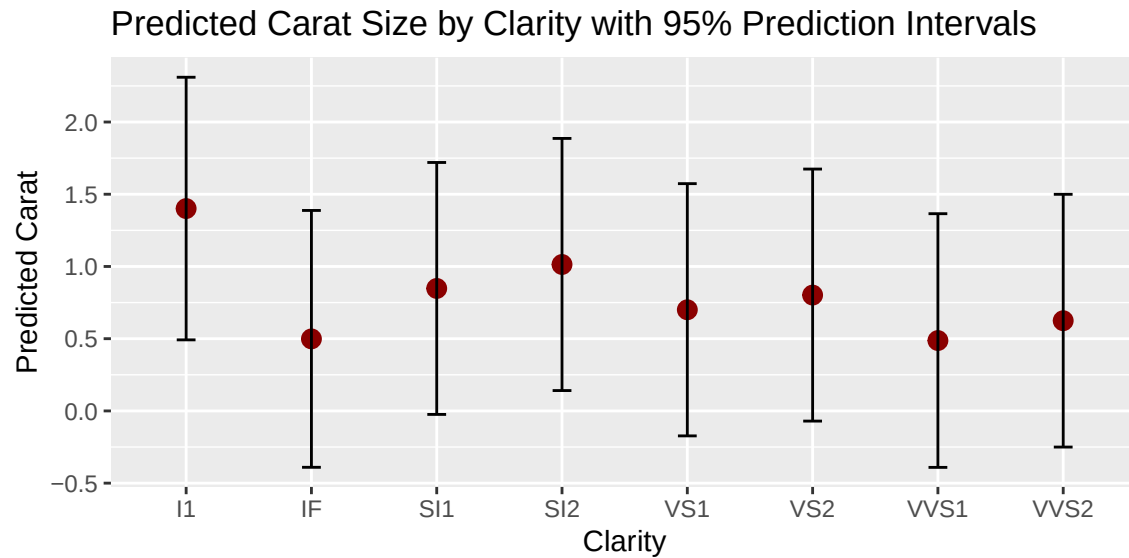


```

pred_df_PI <- cbind(
  clarity_levels,
  predict(simple_model1, newdata = clarity_levels,
          interval = "prediction")
)

ggplot(pred_df_PI, aes(x = clarity, y = fit)) +
  geom_point(size = 3, color = "darkred") +
  geom_errorbar(aes(ymin = lwr, ymax = upr), width = 0.15) +
  labs(
    title = "Predicted Carat Size by Clarity with 95% Prediction Intervals",
    x = "Clarity",
    y = "Predicted Carat"
  )

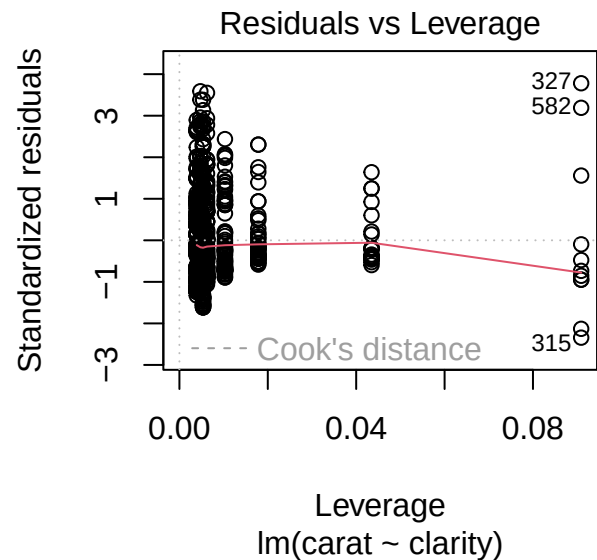
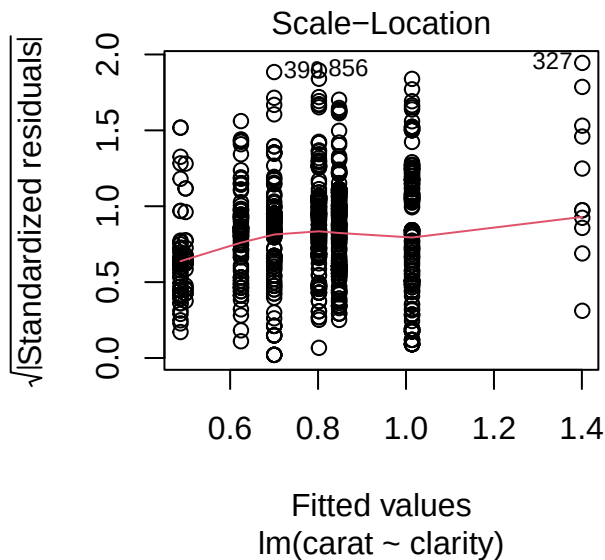
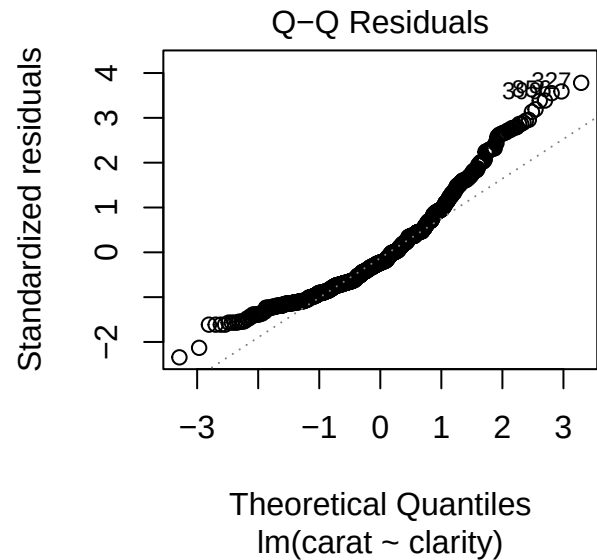
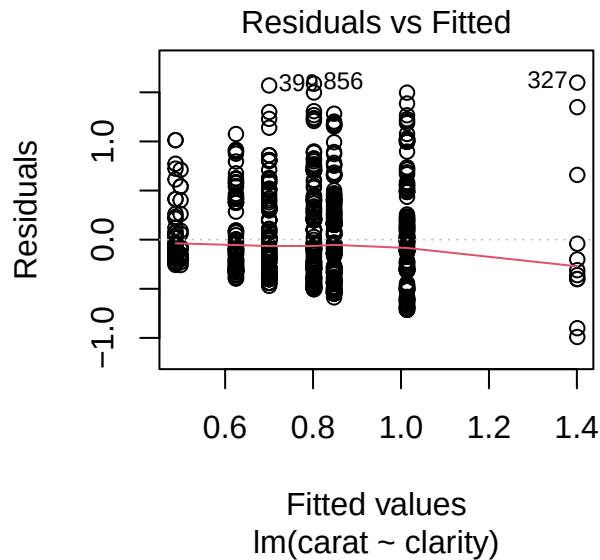
```



The overall significance test for the simple linear regression model reports a p-value of $2.2e-16$, which is far below the 0.05 threshold. This indicates strong evidence that the predictor—clarity—has a statistically significant effect on the carat measurement of a diamond. Similarly, the partial significance test confirms that this predictor is individually significant in explaining variation in the response variable. The adjusted R^2 is nearly identical to the regular R^2 , suggesting that the model is not being penalized for overfitting. However, the R^2 value indicates that the model explains only about 11.45% of the total variability in carat. This relatively low proportion suggests that the simple linear regression model is not capturing much of the underlying structure in the data, and that a more complex or differently specified model may provide a better fit. It's also important to point out that the prediction intervals are wider, as expected, because it's trying to capture an exact carat amount instead of the average carat amount in the confidence interval. In addition, the length of the prediction interval, mathematically speaking, is always wider than the confidence interval of an arbitrary model.

3. Testing Assumptions and Transformations

```
plot(simple_model1)
```

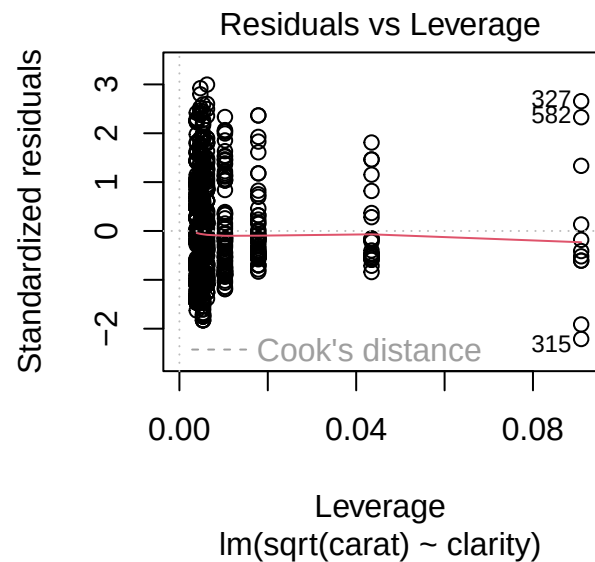
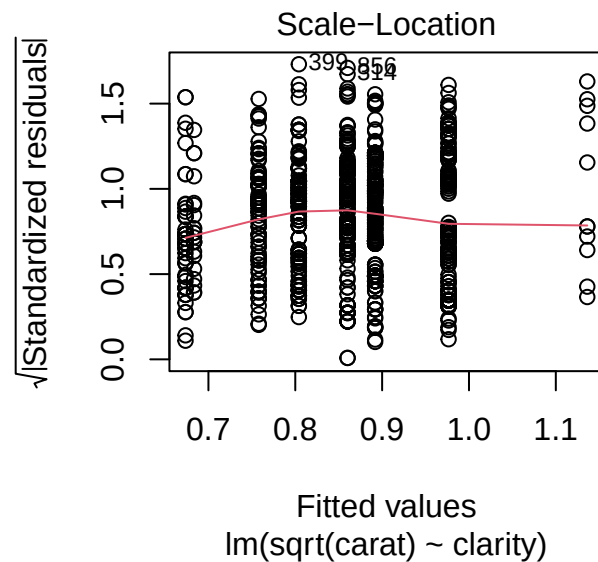
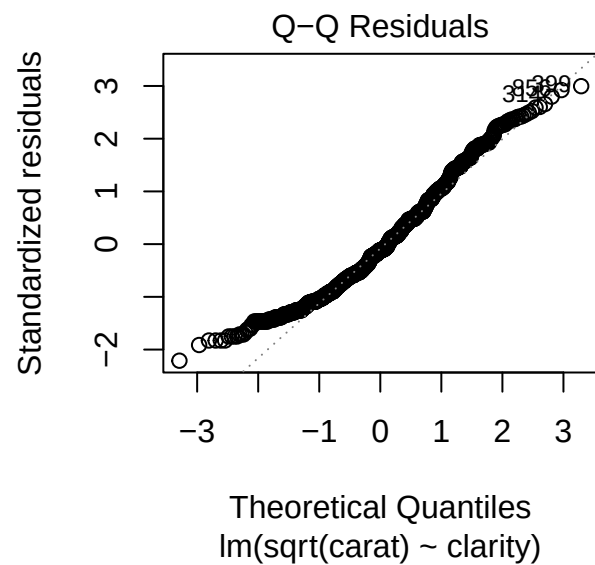
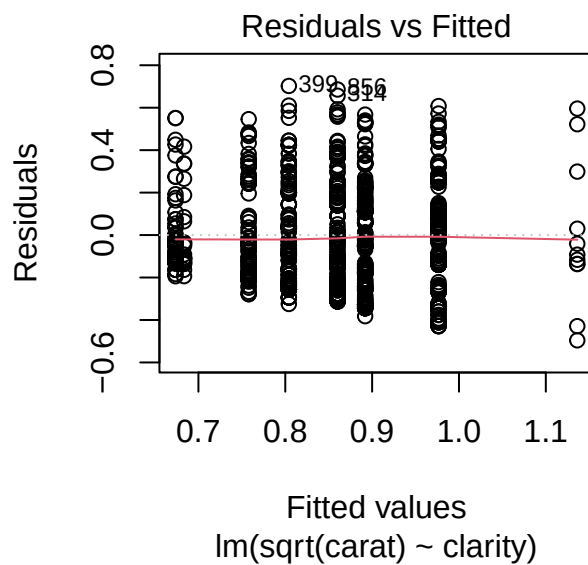


An inspection of the diagnostic plots for the model $\text{carat} \sim \text{clarity}$ shows clear violations of several regression assumptions. The Residuals vs. Fitted and Scale-Location plots both indicate some heteroscedasticity, with a spread that kinda resembles a funnel. The Normal Q-Q plot shows strong deviations in both tails, suggesting that the residuals are not normally distributed. Although the Residuals vs. Leverage plot identifies a few moderately influential points, none appear severely problematic. Overall, the model does not meet the assumptions of normality or constant variance, and a transformation of carat or a more comprehensive model may provide a better fit.

```
sqrt_simple_model1 <- lm(sqrt(carat)~clarity, data= diamond_subset)
sqrt_simple_model1
```

```
##
## Call:
## lm(formula = sqrt(carat) ~ clarity, data = diamond_subset)
##
## Coefficients:
## (Intercept)      clarityIF      claritySI1      claritySI2      clarityVS1      clarityVS2
##      1.1362       -0.4528       -0.2443       -0.1597       -0.3321       -0.2760
## clarityVVS1      clarityVVS2
##     -0.4626       -0.3783
```

```
plot(sqrt_simple_model1)
```



Applying a square-root transformation to the response variable improved the model's adherence to the regression assumptions. The Q-Q plot became noticeably more linear, indicating that the residuals more closely follow a normal distribution. Additionally, the Residuals vs. Fitted and Scale-Location plots showed a more uniform spread of residuals, suggesting that the transformation effectively reduced heteroscedasticity and stabilized the variance across fitted values.

4. Summary on the Square Root Model vs Regular

```
summary(simple_model1)
```

```
##
## Call:
## lm(formula = carat ~ clarity, data = diamond_subset)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.9909 -0.3184 -0.1020  0.2080  1.5991
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.4009     0.1337  10.476 < 2e-16 ***
## clarityIF     -0.9022     0.1626  -5.549 3.69e-08 ***
## claritySI1    -0.5531     0.1366  -4.049 5.54e-05 ***
## claritySI2    -0.3872     0.1376  -2.814 0.00499 **
## clarityVS1    -0.7007     0.1383  -5.068 4.78e-07 ***
## clarityVS2    -0.5989     0.1372  -4.366 1.40e-05 ***
## clarityVVS1   -0.9138     0.1463  -6.247 6.20e-10 ***
## clarityVVS2   -0.7763     0.1411  -5.501 4.80e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4435 on 992 degrees of freedom
## Multiple R-squared:  0.1145, Adjusted R-squared:  0.1083
## F-statistic: 18.33 on 7 and 992 DF,  p-value: < 2.2e-16
```

```
summary(sqrt_simple_model1)
```

```
##
## Call:
## lm(formula = sqrt(carat) ~ clarity, data = diamond_subset)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.49592 -0.18344 -0.02785  0.14477  0.70254
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.13623    0.07092  16.021 < 2e-16 ***
## clarityIF     -0.45279    0.08623  -5.251 1.85e-07 ***
## claritySI1    -0.24426    0.07244  -3.372 0.000775 ***
## claritySI2    -0.15970    0.07298  -2.188 0.028886 *
## clarityVS1    -0.33211    0.07332  -4.530 6.63e-06 ***
```

```
## clarityVS2 -0.27601    0.07275   -3.794 0.000157 ***
## clarityVVS1 -0.46262    0.07758   -5.963 3.43e-09 ***
## clarityVVS2 -0.37833    0.07484   -5.055 5.11e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2352 on 992 degrees of freedom
## Multiple R-squared:  0.1246, Adjusted R-squared:  0.1184
## F-statistic: 20.17 on 7 and 992 DF,  p-value: < 2.2e-16
```

After applying a square root transformation to carat, the regression output showed several improvements. The residual standard error decreased substantially (0.4435 to 0.2352), indicating a tighter model fit. All clarity coefficients became smaller in magnitude but retained their signs and significance, reflecting the compression caused by the transformation. The R^2 and adjusted R^2 values increased slightly, and the F-statistic grew from 18.33 to 20.17, suggesting that clarity explains a bit more variation under the transformed scale. Overall, the transformed model shows improved fit, reduced residual variance, and better adherence to the regression assumptions.

5.(6) Selecting and commenting on the best Model (Background)

Using all selected predictors to model carat substantially improved the overall model performance. Including these variables reduced the residual standard error and increased the F-statistic, both indicating a stronger model fit. The final model demonstrated high explanatory power, accounting for approximately 90% of the total variability in carat. This suggests that the chosen predictors collectively provide a meaningful and effective explanation of differences in diamond size.

Part 3

1. Finding the Best Model

```
carat_model <- lm(carat ~ ., data=sample_diamonds)
#carat_aic <- step(carat_model, direction = 'both')
#summary(carat_aic)
carat_bic <- step(carat_model, direction = 'both', k = log(nrow(sample_diamonds)))
```

```
## Start:  AIC=-5360.6
## carat ~ X + cut + color + clarity + depth + table + price + x +
##      y + z
##
##           Df Sum of Sq  RSS    AIC
## - cut       4    0.0262 3.9792 -5381.6
## - x         1    0.0130 3.9659 -5364.2
## - table     1    0.0198 3.9728 -5362.5
## - X         1    0.0254 3.9783 -5361.1
## <none>                 3.9530 -5360.6
## - y         1    0.0870 4.0400 -5345.7
## - depth     1    0.1448 4.0977 -5331.5
## - z         1    0.2536 4.2066 -5305.3
## - color     6    0.7866 4.7396 -5220.6
## - clarity   7    0.9487 4.9017 -5193.8
## - price     1    3.8939 7.8469 -4681.9
##
```

```

## Step: AIC=-5381.62
## carat ~ X + color + clarity + depth + table + price + x + y +
##      z
##
##      Df Sum of Sq    RSS    AIC
## - x      1      0.0085 3.9877 -5386.4
## - X      1      0.0248 4.0040 -5382.3
## <none>                3.9792 -5381.6
## - table   1      0.0676 4.0468 -5371.7
## - y      1      0.1068 4.0859 -5362.0
## + cut     4      0.0262 3.9530 -5360.6
## - depth   1      0.1340 4.1132 -5355.4
## - z      1      0.2472 4.2264 -5328.2
## - color    6      0.8006 4.7797 -5239.8
## - clarity  7      0.9958 4.9750 -5206.6
## - price    1      3.9125 7.8917 -4703.8
##
## Step: AIC=-5386.38
## carat ~ X + color + clarity + depth + table + price + y + z
##
##      Df Sum of Sq    RSS    AIC
## - X      1      0.0254 4.0131 -5386.9
## <none>                3.9877 -5386.4
## + x      1      0.0085 3.9792 -5381.6
## - table   1      0.0651 4.0528 -5377.1
## - y      1      0.1041 4.0918 -5367.5
## + cut     4      0.0218 3.9659 -5364.2
## - depth   1      0.1775 4.1652 -5349.7
## - z      1      0.3874 4.3751 -5300.6
## - color    6      0.7971 4.7848 -5245.6
## - clarity  7      0.9880 4.9757 -5213.4
## - price    1      3.9041 7.8918 -4710.7
##
## Step: AIC=-5386.94
## carat ~ color + clarity + depth + table + price + y + z
##
##      Df Sum of Sq    RSS    AIC
## <none>                4.0131 -5386.9
## + X      1      0.0254 3.9877 -5386.4
## + x      1      0.0092 4.0040 -5382.3
## - table   1      0.0620 4.0751 -5378.5
## - y      1      0.1091 4.1222 -5367.0
## + cut     4      0.0213 3.9918 -5364.6
## - depth   1      0.1819 4.1950 -5349.5
## - z      1      0.3920 4.4051 -5300.6
## - color    6      0.7920 4.8051 -5248.3
## - clarity  7      0.9998 5.0129 -5212.8
## - price    1      4.0575 8.0706 -4695.2

```

```
summary(carat_bic)
```

```

##
## Call:
## lm(formula = carat ~ color + clarity + depth + table + price +

```

```
##      y + z, data = sample_diamonds)
##
## Residuals:
##      Min        1Q      Median        3Q        Max
## -0.38405 -0.02894 -0.00384  0.03014  0.65321
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.580e+00  3.870e-01   4.082 4.82e-05 ***
## colorE       1.604e-02  7.649e-03   2.096 0.036294 *
## colorF       1.124e-02  7.666e-03   1.466 0.142947
## colorG       1.112e-02  7.343e-03   1.514 0.130264
## colorH       5.133e-02  7.814e-03   6.569 8.20e-11 ***
## colorI       7.375e-02  8.696e-03   8.481 < 2e-16 ***
## colorJ       1.047e-01  1.057e-02   9.906 < 2e-16 ***
## clarityIF    -3.000e-01  2.480e-02 -12.098 < 2e-16 ***
## claritySI1   -2.592e-01  2.033e-02 -12.749 < 2e-16 ***
## claritySI2   -2.210e-01  2.030e-02 -10.888 < 2e-16 ***
## clarityVS1   -2.817e-01  2.092e-02 -13.460 < 2e-16 ***
## clarityVS2   -2.771e-01  2.067e-02 -13.406 < 2e-16 ***
## clarityVVS1  -2.748e-01  2.240e-02 -12.267 < 2e-16 ***
## clarityVVS2  -2.855e-01  2.153e-02 -13.257 < 2e-16 ***
## depth       -4.009e-02  6.012e-03  -6.668 4.33e-11 ***
## table        3.919e-03  1.006e-03   3.894 0.000105 ***
## price        4.172e-05  1.325e-06  31.494 < 2e-16 ***
## y           -3.053e-01  5.911e-02  -5.165 2.91e-07 ***
## z            9.289e-01  9.489e-02   9.789 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.06396 on 981 degrees of freedom
## Multiple R-squared:  0.9818, Adjusted R-squared:  0.9815
## F-statistic: 2938 on 18 and 981 DF, p-value: < 2.2e-16
```

After conducting model selection using both the AIC and BIC criteria, we found that each method identified the same optimal model. This reduced model retained price, table, color, and clarity as predictors. The resulting fit demonstrated strong performance, with the adjusted R^2_{adj} closely aligning with the regular R^2 , indicating that the model explains the data well without evidence of overfitting. Notably, the model accounted for approximately 98% of the variability in diamond size, suggesting exceptionally high explanatory power. The residual standard error was also low, further supporting the model's precision. Additionally, the large F-statistic provides evidence of strong overall model significance and reinforces confidence in the selected predictors.

2. Detecting Multicollinearity (+ Testing of Assumptions + Transformations)

```
library(car)
```

```
## Loading required package: carData
```

```
##
```

```
## Attaching package: 'car'
```

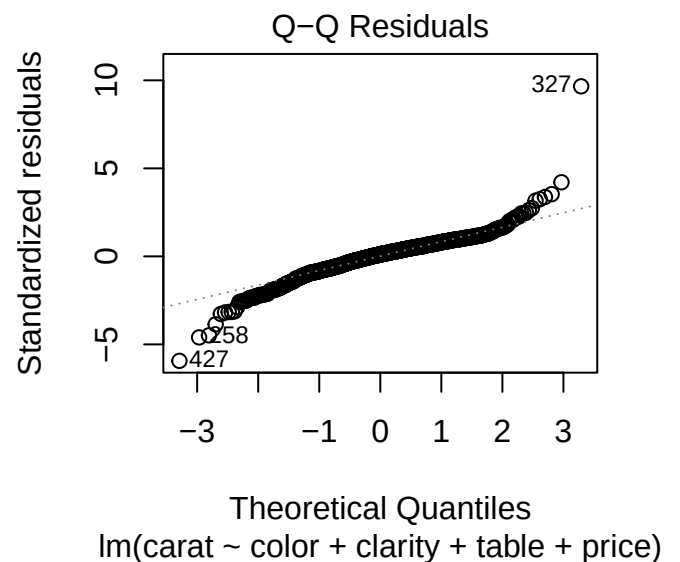
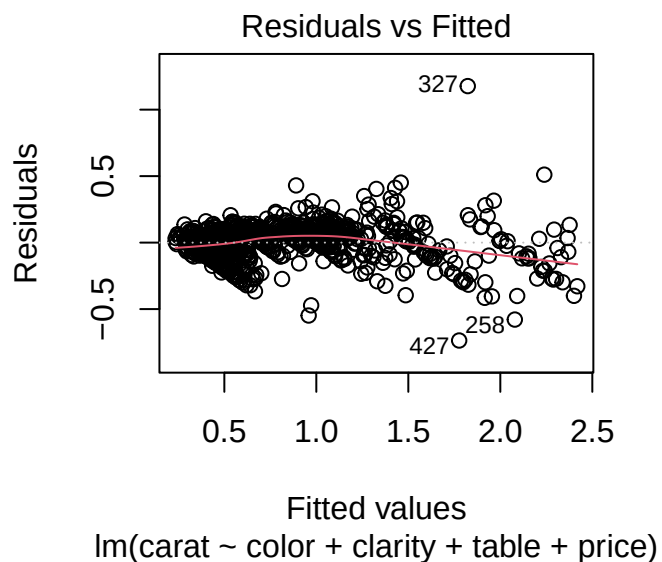
```
## The following object is masked from 'package:dplyr':
##
##   recode
```

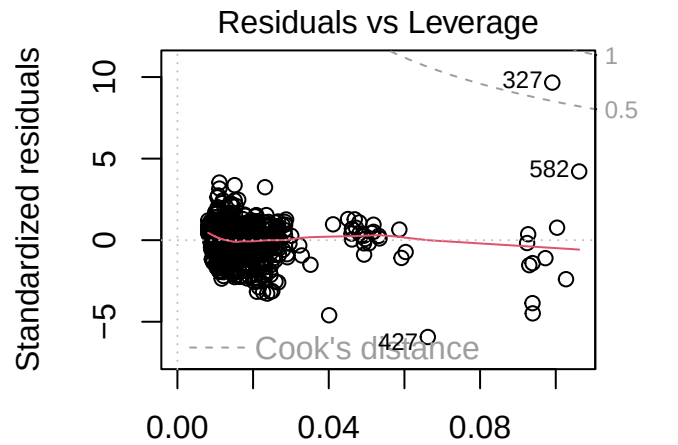
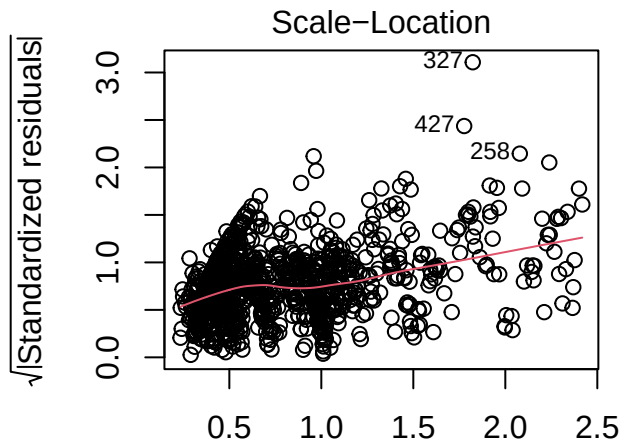
```
## The following object is masked from 'package:purrr':
##
##   some
```

```
vif(carat_bic)
```

```
##           GVIF Df GVIF^(1/(2*Df))
## color      1.270936 6      1.020180
## clarity    1.588815 7      1.033624
## depth     16.997671 1      4.122823
## table      1.159087 1      1.076609
## price      6.476474 1      2.544892
## y         1037.824048 1     32.215277
## z         1024.647428 1     32.010114
```

```
carat_vif <- lm(
  carat ~ color + clarity + table + price, data=sample_diamonds
)
plot(carat_vif)
```





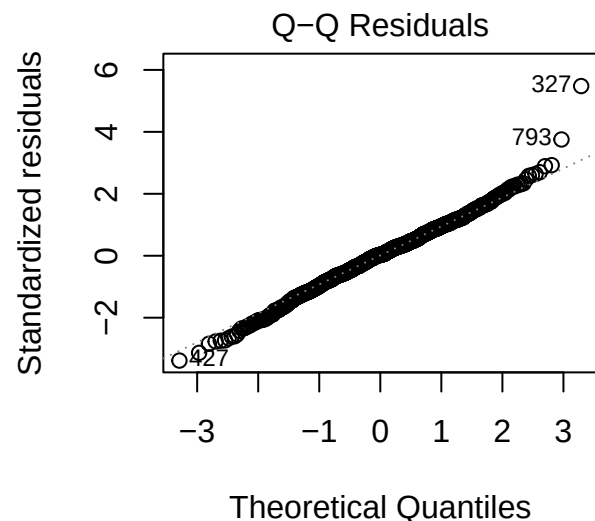
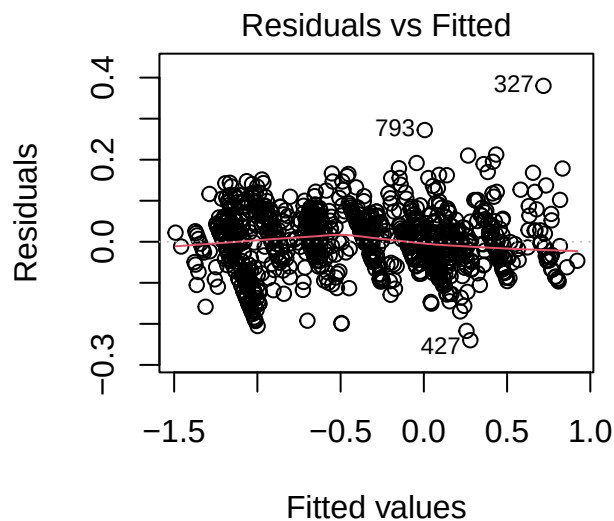
```
# Checking Assumptions before Transforming!!!
transformed_carat_vif <- lm(
  log(carat) ~ color + clarity +
  log(table) + log(price), data=sample_diamonds)

summary(transformed_carat_vif)
```

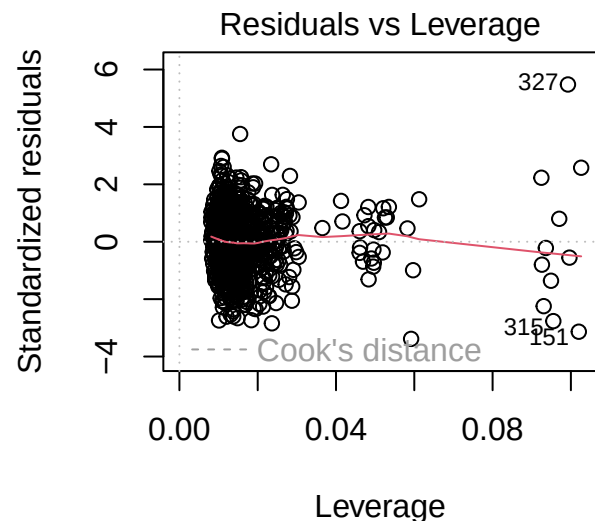
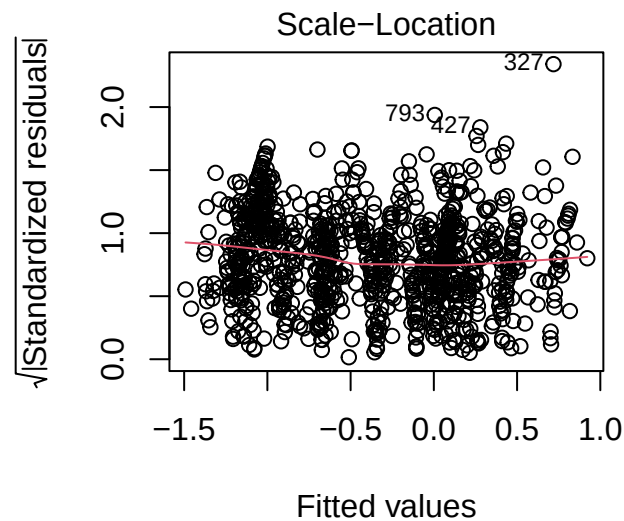
```
##
## Call:
## lm(formula = log(carat) ~ color + clarity + log(table) + log(price),
##     data = sample_diamonds)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.23992 -0.04491  0.00223  0.04656  0.38000
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -5.197254   0.256829 -20.236  < 2e-16 ***
## colorE       0.051469   0.008681   5.929 4.22e-09 ***
## colorF       0.065292   0.008720   7.488 1.56e-13 ***
## colorG       0.091588   0.008322  11.006  < 2e-16 ***
## colorH       0.143510   0.008775  16.354  < 2e-16 ***
## colorI       0.204050   0.009736  20.959  < 2e-16 ***
## colorJ       0.294225   0.011540  25.497  < 2e-16 ***
## clarityIF    -0.641106   0.026923 -23.813  < 2e-16 ***
## claritySI1   -0.381923   0.022634 -16.874  < 2e-16 ***
## claritySI2   -0.297797   0.022795 -13.064  < 2e-16 ***
## clarityVS1   -0.498540   0.022876 -21.793  < 2e-16 ***
## clarityVS2   -0.456914   0.022734 -20.098  < 2e-16 ***
## clarityVVS1 -0.611234   0.024203 -25.255  < 2e-16 ***
## clarityVVS2 -0.570620   0.023370 -24.417  < 2e-16 ***
```

```
## log(table)    0.261151    0.063466    4.115 4.20e-05 ***
## log(price)    0.524483    0.002390 219.407 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.07308 on 984 degrees of freedom
## Multiple R-squared:  0.9843, Adjusted R-squared:  0.984
## F-statistic: 4102 on 15 and 984 DF,  p-value: < 2.2e-16
```

```
plot(transformed_carat_vif)
```



$\text{lm}(\log(\text{carat}) \sim \text{color} + \text{clarity} + \log(\text{table}) + \log(\text{price}))$



$\text{lm}(\log(\text{carat}) \sim \text{color} + \text{clarity} + \log(\text{table}) + \log(\text{price}))$

Due to the substantial multicollinearity among the predictors x, y, and z in the original model, we elected to use a revised model that excludes these variables. After their removal, the remaining predictors exhibited VIF values close to 1, indicating negligible multicollinearity. Although this decision resulted in a modest reduction in the model's R^2 , the revised model still accounted for 93% of the total variability in the response, suggesting that explanatory power remained strong. The residual standard error increased roughly to 12%, reflecting a minor decline in predictive accuracy. Nonetheless, we selected this reduced model over the original because it offers a more interpretable structure and clearer relationships between the predictors and the response variable. In the end—when we tested the assumptions on the model—we found that the quantitative predictors actually had a log linear relationship toward the response variable carat. This is due to the fact that after we transformed the model all the assumptions (normality, homoscedasticity, power of influencers) of the LRM were now followed by the model.

3. Prediction and Confidence Intervals of a Unique Combination of Predictors

```
X_values <- data.frame(color = "F", clarity = "SI1", price = 10000, table = 54)
predict(transformed_carat_vif, X_values, interval = "confidence")
```

```
##           fit          lwr          upr
## 1 0.3585082 0.3410483 0.3759681
```

```
predict(transformed_carat_vif, X_values, interval = "prediction")
```

```
##           fit          lwr          upr
## 1 0.3585082 0.2140364 0.5029799
```

4. CONCLUSION

This project analyzed a random sample of 1,000 diamonds to examine the relationships among both quantitative and categorical variables with the goal of building regression models for prediction and inference. The simple linear regression focused on predicting initial price using individual predictors, while the multiple regression model concentrated on explaining variation in diamond size (carat). Initial exploratory analysis revealed several skewed distributions, suggesting the need for potential transformations to better satisfy model assumptions. Tools such as ggpairs further supported informed variable selection by illustrating associations among predictors and the response.

The simple linear regression using clarity as the sole predictor demonstrated limited explanatory power, and diagnostic plots indicated violations of normality and homoscedasticity. These issues improved substantially after applying a square-root transformation to the response variable.

For the multiple linear regression model, AIC and BIC criteria were used to identify an effective set of predictors. We also cut three predictors from the model using VIF calculations for finding multicollinearity, to make the model more interpretable while slightly losing a bit in prediction power. Additional logarithmic transformations were applied to ensure that the final model satisfied key assumptions of normality, homogeneity of variance, and resistance to the influence of outliers.